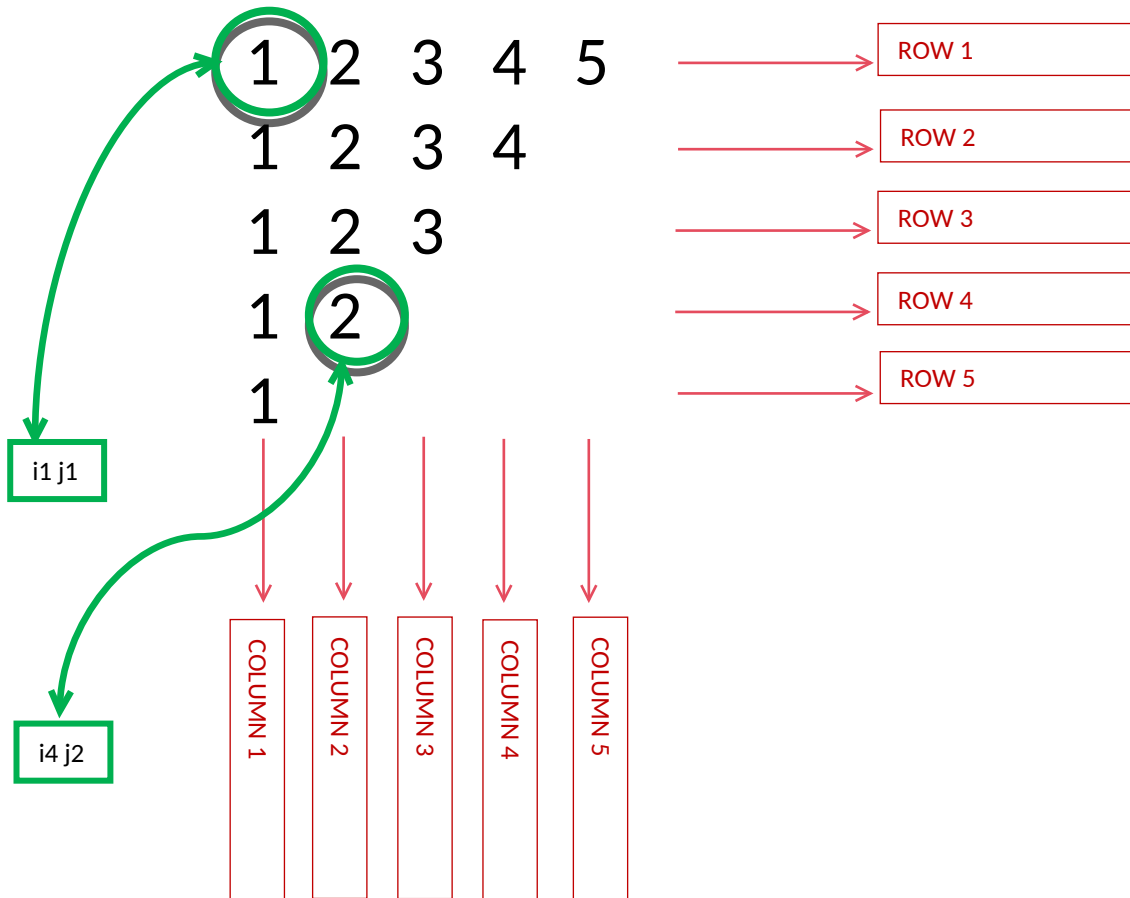


WEEK 01 MODULE
Programming Practice

TOPIC : 2 'Number Pattern'

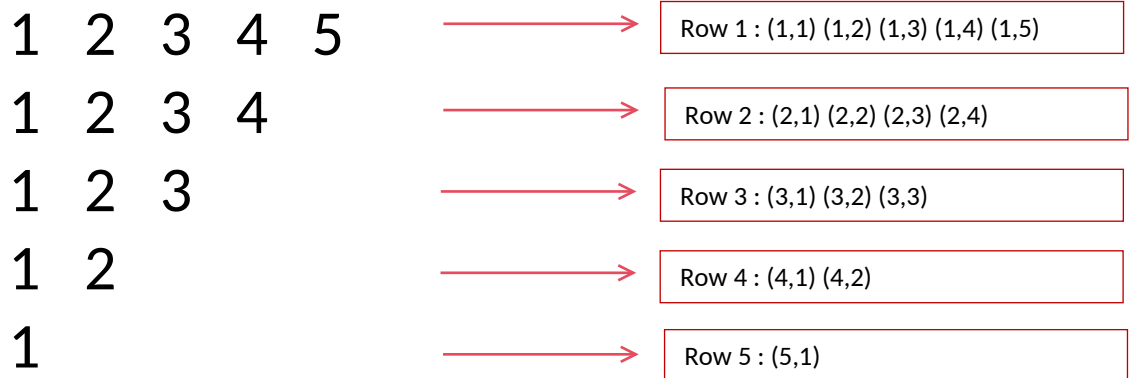
5. INVERTED NUMBER TRIANGLE



1. Problem Analysis :-

i. Pattern Structure:

- The output forms an inverted numbered right-angled triangle.
- This pattern is also based on the **Right Angled Triangle structure**, but inverted.
- Unlike the Number Triangle, here the number of elements **decreases row by row**.
- The first row contains all numbers from 1 to n, and each subsequent row reduces by one element.
- For n=5



ii. Execution Flow:

Step 1: Observe the Grid Concept:

- The pattern can be visualized as a two-dimensional grid (matrix)
- size = $n \times n$.
- Each cell in the grid represents a unique coordinate (i, j), where i denotes the row index and j denotes the column index.
- For $n = 5$:

Observation:

Row 1 → 1 2 3 4 5
 Row 2 → 1 2 3 4
 Row 3 → 1 2 3
 Row 4 → 1 2
 Row 5 → 1

A clear relationship can be observed:

- First row has n elements
- Each row reduces by 1 element
- Values still start from 1 in every row

Step 2: Traverse the grid row by row using outer loop.

- The outer loop is responsible for generating rows.
- for (i = 1 to n)

Step 3: For every row, number of elements decreases .

- for (j = 1 to (n-i+1))

Step 4: Print value = j

Step 5: After all numbers of the current row have been printed, move the cursor to the next line.

Step 6: Repeat the same process until all n rows have been completed and pattern shrinks to 1 element

2. Program :-

```
package week1_module;
import java.util.Scanner;
public class InvertedNumberTraingle {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of rows: ");
        int n = sc.nextInt();

        for(int i = 1; i <= n; i++) {

            for(int j = 1; j <= (n - i + 1); j++) {
                System.out.print(j + " ");
            }

            System.out.println();
        }

        sc.close();
    }
}
```

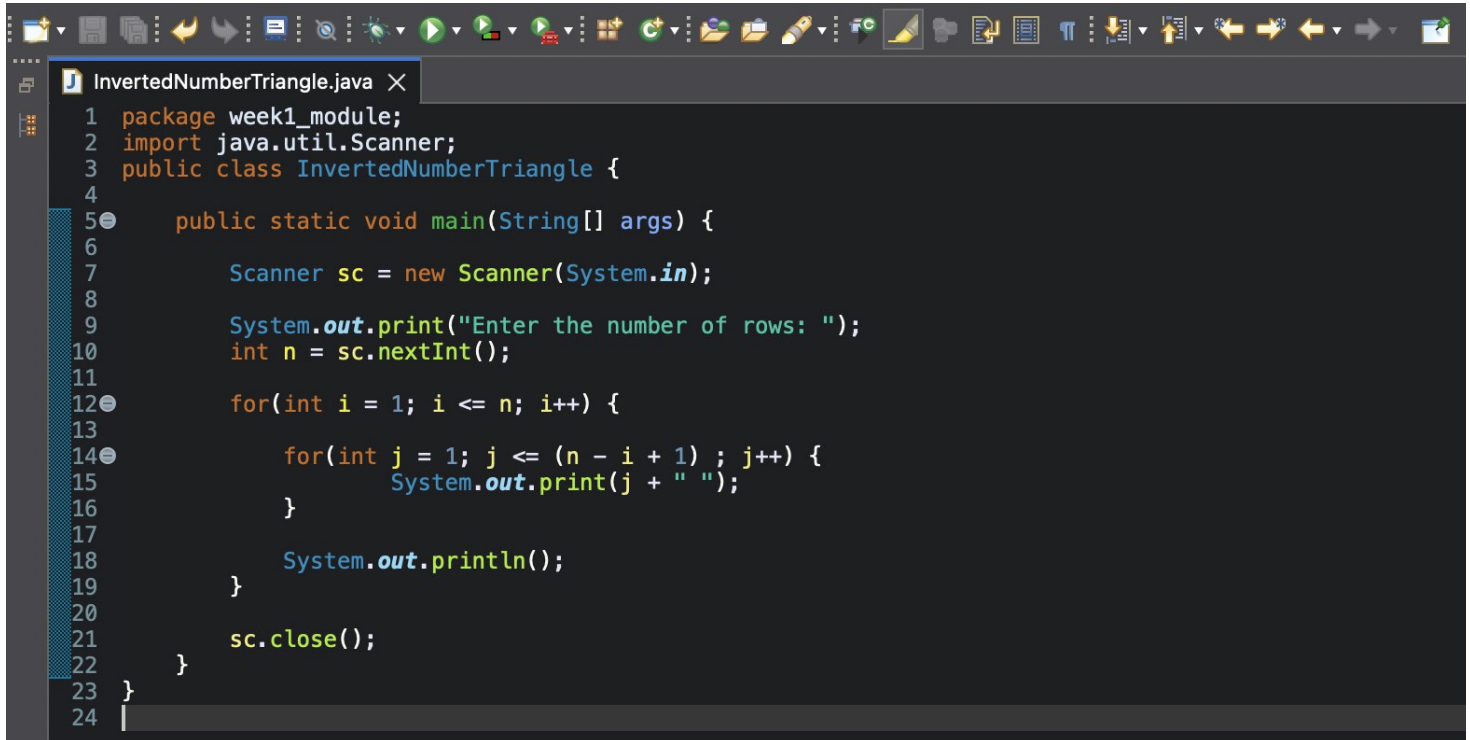
3. Program Output :-

Enter the number of rows: 5

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

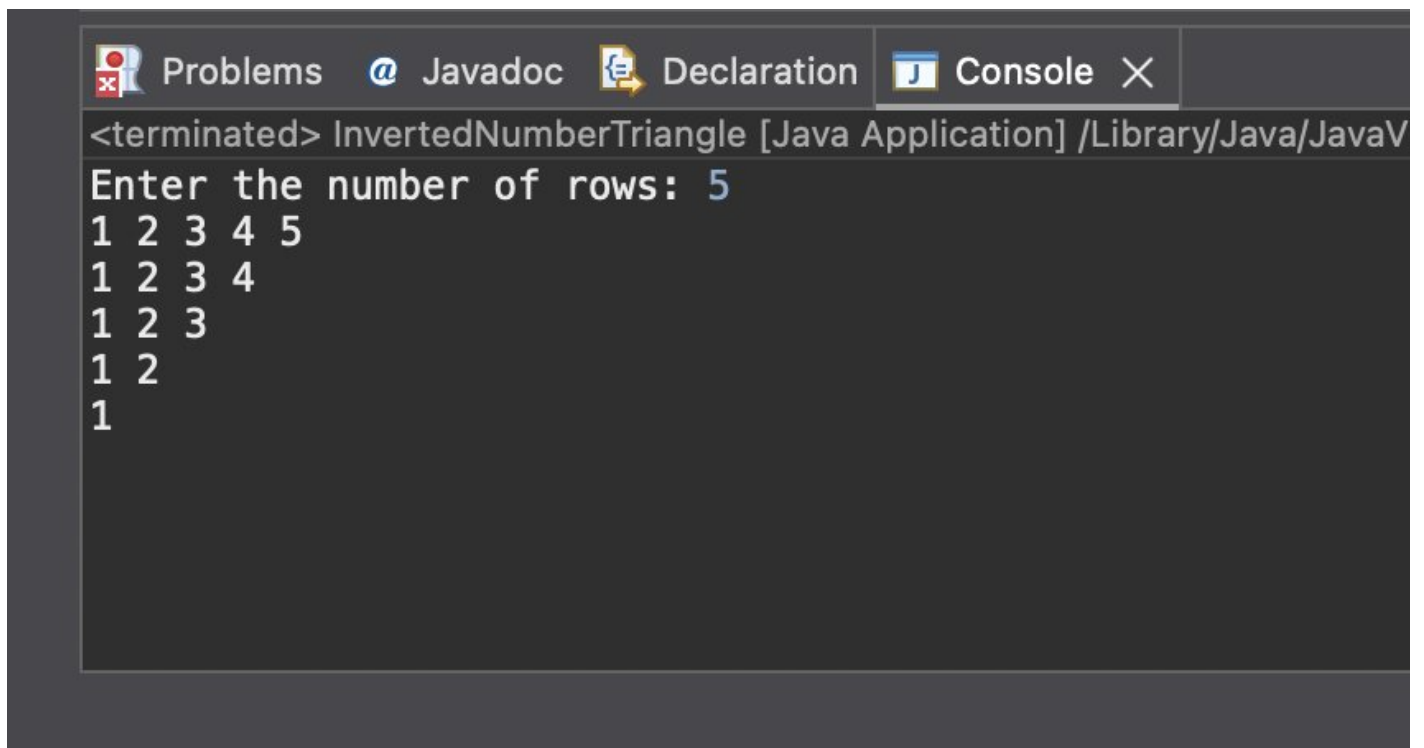
4. ScreenShot :-

i. Program :-



```
1 package week1_module;
2 import java.util.Scanner;
3 public class InvertedNumberTriangle {
4
5     public static void main(String[] args) {
6
7         Scanner sc = new Scanner(System.in);
8
9         System.out.print("Enter the number of rows: ");
10        int n = sc.nextInt();
11
12        for(int i = 1; i <= n; i++) {
13
14            for(int j = 1; j <= (n - i + 1); j++) {
15                System.out.print(j + " ");
16            }
17
18            System.out.println();
19        }
20
21        sc.close();
22    }
23 }
24
```

ii. Program's Output :-



```
<terminated> InvertedNumberTriangle [Java Application] /Library/Java/JavaV
Enter the number of rows: 5
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```